

BOHAN BAI

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EDUCATION

DUKE UNIVERSITY, The Fuqua School of Business

Durham, NC

Master of Science in Quantitative Management (Business Analytics), Finance Track

May 2025

Data Science Coursework: Data Science for Business, Decision Analytics & Modeling, Data Visualization, Data infrastructures

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, CA

Bachelor of Arts in Economics & Minor in Computer Science

Sep 2021 – Jun 2024

Publication: How Marriage Policy Affects Women's Labor Force Participation, Bohan Bai, ICGPSH 2023

RESEARCH EXPERIENCE

Artificial Intelligence Applications in Development Economics

Zurich

Pre-Doctoral Researcher, Supervisors: Prof. David Yanagizawa-Drott

Sep 2025 – Present

- Build multi-agent LLM "AI scientist" systems that design and iterate field-experiment nudges.
- Estimate causal effects (ITT/OLS, robust SEs, CATE) to benchmark AI vs. human treatments.
- Develop LLM-simulated personas and embedding + gradient-boosting surrogate models of behavior.
- Build self-improving chatbots optimized via contextual bandits and prompt search.

Clifton Strengths Data Assistant–Duke Pratt School of Engineering

Durham

Data analyst, Supervisors: Cameo Hartz

Nov 2024 – Present

- Designed and standardized a data collection strategy to simplify user operations.
- Analyzed datasets to identify important trends and created explorative data visualizations
- Proposed machine learning and statistical models to assess and optimize behavioral impacts.

Duke Bass Connection Program–Scholar Academy for Latinxs United for Diversity (SALUD)

Durham

Data analyst/project manager, Supervisors: Scarlet Soriano

Sep 2024 – Present

- Developed multiple data visualizations for Census data on High school teachers in Uganda.
- Perform Data Analysis with Program available data to analyze the effectiveness of the program
- Training students in relevant project management and research methodologies
- Meeting weekly with the project leaders to discuss team progress and next steps

Undergraduate Research Association Program (URAP)

Developed multiple data visualizations for Census data on High school teachers in Uganda

Berkeley

Supervisors: Frederico S Finan, Silvia Barbareschi

Jan 2024 – May 2024

- Implemented various regression methods on the Census dataset to analyze relationships between Variables. Assisted in writing the results report of data analysis and addressing the research question

Employee selection tendencies of large and small enterprises in the market

Berkeley

Supervisors: Demian Pouzo, Ross Chu, Tammy Lee

Jan 2023 – Oct 2023

- Learned to use the SCRAPY crawler framework, able to crawl over 100,000 data at once
- Conducted basic analysis on the crawling data using NumPy and Pandas and assisted in writing the results report of data analysis.

Econ 191 Topics in Economics Research

Berkeley

The impact of marriage policies on the female labor force

Jan 2023 – Oct 2023

Supervisors: Benjamin R. Handel, Alfredo Mendoza Fernandez, Petra Laura Oreskovic

- Analyzed large data sets retrieved from PSID by using Pandas and Numpy
- Applied the DID regression analysis method with data
- Completed a review paper (submitted) after conducting thorough literature review

Distributional Environmental Justice On Santa Barbara County

Berkeley

The impact of pesticide usage distributes in Santa Barbara County

Sep 2021 – June 2022

Supervisor: Dana Montague

- Collected data and conducted analysis of data using spatial regression analysis
- Completed a comprehensive research report on the analysis results

TECHNICAL CAPABILITIES

Java, Python, R, C++, SQL, Stata, Scrapy, LaTeX, MATLAB, Microsoft Offices, Adobe Products, word press, Wix, IDEs, GitHub, Google Code Lab, Jupiter Notebook